

I am currently a fifth year PhD student in OR at UC, Irvine. I am looking for an internship for 2026 summer. I am a data-driven Operations Researcher specializing in optimization for complex decision systems.

RESEARCH EXPERIENCE

Enrollment Prediction and Admission Decisions Optimization

OCT 2022-PRESENT

Co-author | Machine Learning & Reinforcement Learning

- Designed an **adaptive Predict-Then-Optimize framework** that iteratively updates prediction accuracy and system states based on real-time decision outcomes, improving **sequential decision-making** under uncertainty.
- Applied approximate dynamic programming algorithms integrating **reinforcement learning** and **mathematical programming** to model **uncertain** admitted-applicant behavior, **stochastic** demand, and operational constraints, enhancing revenue management and achieving target class size and composition.
- Engineered full **data processing pipeline** in python (missing data handling, one-hot encoding, feature engineering, outlier treatment), improving data quality for downstream predictive modeling and optimization.
- Built and compared two **predictive** modeling frameworks for enrollment **likelihood forecasting** (direct vs. indirect probability modeling) using **machine learning** algorithms such as **XGBoost** and **Logistic Regression**; the indirect framework reduced false positive predictions by 23% and improved true positives by 27%.
- Conducted **causal and behavioral analysis** on the impact of scholarship levels on applicant enrollment decisions using propensity score matching to reduce selection bias, informing financial aid allocation strategy.
- Developed and evaluated multiple admission optimization algorithms, including static **Gurobi-based** models, **simulation-based** policy evaluation, and **reinforcement learning** with approximate value functions.

Corn Planting and Harvest Scheduling Considering Storage Capacity and Daily Temperature

MAR 2021-SEP 2022

Co-author | Machine Learning & Deep Learning

- Formulated **large-scale** planting and harvest **scheduling problems** for 1,375+ corn populations across two locations using Mixed-Integer Linear Programming (MILP), balancing weekly harvests under storage capacity constraints.
- Preprocessed large-scale agricultural and environmental data in python through data cleaning, feature engineering, and time-series preparation, including applying moving averages to smooth temporal variations.
- Implemented and compared **Machine Learning and Deep Learning forecasting models**, such as LSTM (PyTorch), SARIMA (R), Prophet, CNN, and ensemble models for daily temperature forecasting using 10 years of historical data; validated models via k-fold cross-validation, achieving RMSE = 0.28 and 95% correlation in harvest date prediction.
- Optimized the planting-harvest schedule using the **Gurobi solver and simulation**, producing 51 balanced harvest weeks compared to the original schedule with 35 under-capacity and 17 over-capacity weeks.

Time-varying mediation analysis for incomplete data with application to DNA study for PTSD

MAY 2022-DEC 2022

Research Assistant | Longitudinal Data, Missing Data, Multiple Imputation

- Conducted exploratory data analysis in R on **high-dimensional longitudinal mediators** involving 1,879 DNA methylation biomarkers measured across 4 time points for 526 subjects, including statistical summaries, temporal pattern visualization, correlation structure analysis, and data quality assessment.
- Addressed complex missing data affecting 58.3% of subjects with non-monotone missingness by using LMI method.

CONFERENCE PRESENTATION

INFORMS: Presented the paper "*Admit or Wait: Dynamic Admission Control under Non-Stationary Arrivals*" at **INFORMS Annual Meetings (2024, 2025)** and the **DSI Annual Conference (2025)**. The paper is ready for submission to Operations Research, a top journal in my area.

EDUCATION

UNIVERSITY OF CALIFORNIA, IRVINE

- PhD in Operations and Decision Technologies | Paul Merage Business School SEP 2021-PRESENT
- Relevant Coursework: Algorithm Design | Large Optimization | Dynamic Programming | Applied Econometrics
- Reward: Ray Watson Doctoral Fellowship

WASHINGTON UNIVERSITY IN ST. LOUIS

JUL 2019-JAN 2021

- Master of Science in Supply Chain Management | Olin Business School
- Relevant Coursework: Operations Management | Machine Learning | Deep Learning | SQL | Big Data Computing

TIANJIN NORMAL UNIVERSITY, CHINA

SEP 2013-JUN 2017

- Bachelor of Science in Logistics Management | Business School